

FLEA NEWS 25

Department of Entomology
Iowa State University
Ames, Iowa 50011 U. S. A.

November 1982

Compiled by Robert E. and Joanne H. Lewis

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Please find herewith:

- lists of additional literature for 1973 (list 13), 1974 (list 14), 1975 (list 15), 1976 (list 13), 1977 (list 11), 1978 (list 12), 1979 (list 11), 1980 (list 7), 1981 (list 4) and 1982 (list 2)
- additions, changes and deletions to the mailing list

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RECENT COMMUNICATION with various colleagues suggest that 1982 may see the publication of two major works on the flea fauna of the world. Dr. ROBERT TRAUB and his assistant, Ms HELLE STARCKE, report progress on the ceratophyllid volume of the Catalogue series and predict its publication shortly. Dr. LI KUEI-CHEN, Guiyang, P.R.C., and Dr. LIU CHI-YING, Beijing, P.R.C., are completing the manuscript of a Monograph of the Fleas of China, publication date as yet unknown. In addition, Dr. CLUFF HOPLA, Norman, Oklahoma, writes that he is about 3/4 finished with the illustrations for his work on the Siphonaptera of Oklahoma and expects it to be published some time in 1983. The editors would appreciate hearing about other major projects that are currently underway.

Mr. STEPHEN SMITH, Columbus, Ohio, writes that he is suspending his studies of flea larvae while he pursues his D.V.M. at the Ohio State University School of Veterinary Medicine.

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DELETIONS

Mr. L. S. Cuy [mail returned marked deceased]

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RECENTLY PUBLISHED - The Rodents of Southern Africa by G. De Graaff, 1981
(ISBN 0 409 09829 0) Butterworth & Co. (SA) (PTY) Ltd., 152-154 Gale Street,
Durban 4001, S.A. i-xviii, 1-267, 12 colour plates, 111 figures, 73 maps.
U.S. \$42.53.

While the large mammals of Africa traditionally have attracted attention from sportsmen and scientists alike, the same cannot be said for the small mammals. Indeed, many faunal elements are extremely poorly known and additional new taxa still are being described annually. Although the identification manual edited by Meester and Setzer (1971) gave stability to the nomenclature, there has long been a need for an in-depth treatment of the African fauna which could be used by the non-specialist. The Rodents of Southern Africa is such a book.

As stated in the Preface, the book is not intended to be a systematic treatise, but rather a book that can be used in teaching classes dealing with mammals and as a reference work for the layman or field worker who wishes to learn something about interesting adaptations and ways of life of a very important order of mammals. While subspecies are mentioned occasionally in the text, the basic taxonomic unit is the species. Though the author assumes that the reader will be familiar with the basic principles of general zoology and its related terminology, the introduction includes considerations of such topics as dental formulae, zoogeographic realms, skeletal features and terminology, organ systems, etc. This is followed by a comparison of the Wood (1955) and Anderson (1967) systems of rodent classification and a checklist of southern African rodents.

The bulk of the text consists of the species accounts, in which the information is arranged in a uniform format for each taxon. Following the generic and specific names (including the author and date of description) are the common names in English and Afrikaans. A brief outline of the synonymic history comes next, followed by a tabular display of the body measurements for both sexes. The remainder of the account is divided by the following headings: identification, skull and dentition (including photographs of the skull in dorsal, ventral, and lateral view, and an enlargement of the cheekteeth), distribution, habitat, diet, habits, predators, reproduction, parasites (both internal and external), relations with man, prehistory and taxonomy. Each account is also furnished with a distribution map, and sometimes line drawings. Twelve attractive coloured plates depicting the species in their normal habitat are included. The book concludes with a glossary, a list of references, and an index.

One attractive innovation to be found in this volume involves the keys. Each of these is accented by being printed over a light gray background, as are the tables and line drawings. The result is pleasing to the eye and greatly facilitates locating the keys in the various sections.

The text is remarkably free of mechanical flaws, printed in a pleasing format, and well reflects the obvious affection of the author for his subject matter. In these days of inflated book prices, the cost is modest for a tome of this quality.
--R.E.L.

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RECENTLY PUBLISHED - Mammal Species of the World: a Taxonomic and Geographic Reference edited by J. H. Hunacki, K. E. Kinman and J. W. Koepl, 1982 (ISBN 0 942924 00 2). Published as a joint venture of The Allen Press, Inc. and The Association of Systematics Collections (JVA-Mammals, P.O. Box 368, Lawrence, Kansas 66044, USA). 694 pp. U.S. \$55.00.

While no mention is made in this publication of mammal parasites, the volume is an invaluable reference to parasitologists in general. It embodies the contributions of 189 specialists in mammalian taxonomy from 23 countries and treats 4,170 currently recognized species.

The editors frequently refer to this volume as a checklist in their introduction, but it more accurately fulfills the definition of a catalogue of the mammalian fauna of the world. The taxonomic arrangement follows that of Corbet and Hill (1980, A World List of Mammalian Species) with a few minor modifications. Entries are listed phylogenetically by order and family, and alphabetically by genus and species. A typical species treatment includes the following information: current generic and specific name, author, date and citation of original description, names of reviewers, type locality, and known distribution. This information frequently is followed by a comments section which includes information on synonymic history and treatment by other specialists. Most treatments conclude with the International Species Inventory System number. Those species governed by the regulations of the U.S. Endangered Species Act and those listed in the appendices of the Convention on International Trade in Endangered Species of Wild Fauna and Flora are also annotated under a Protected Status heading. The index includes all taxonomic names included in the volume, including synonyms, nomina nuda, nomina oblita, and other invalid names. There are also 15 pages of bibliography and literature cited.

While this volume does little toward sorting out the subspecies confusion so typical of mammalian taxonomy, it certainly provides a major resource to those

who concern themselves with mammalian associates. The editors, contributors, and reviewers are to be congratulated for their perseverance. --R.E.L.

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FLEA NEWS AND EDITOR, R. LEWIS, ACHIEVE NATIONAL RECOGNITION IN U.S.

In the Spring of 1982, pulicology sprang from the laboratory into the lime-light for a few months. It all began when Mr. Robert Hullihan published in the Des Moines [Iowa] REGISTER a series of interviews with a number of Iowa citizens which he entitled "Focus on a Foremost Authority".* One of these interviews was with Prof. Robert Lewis and discussed his work on fleas, publication of FLEA NEWS, and Prof. Lewis's aspirations to build a data bank of Siphonaptera information with computer resources. The ink on the newsprint was scarcely dry when a reporter from the WALL STREET JOURNAL, a prestigious financial newspaper with national circulation, telephoned to request an interview. Ms Sue Shellenbarger arrived on April 1, and the fruits of her pen appeared May 13 on page 1 of the WALL STREET JOURNAL. The article is reprinted in this issue of FLEA NEWS. Ms Shellenbarger also consulted with leading authorities R. Traub, M. Rothschild, and W. L. Jellison. In addition to those mentioned in the article, we understand she also consulted with C. Hopla and N. Gratz. Her article succeeded very well in presenting the state of the science in pulicology.

Immediately upon publication of the article, Prof. Lewis was flooded with telephone calls from friends, radio stations, and people with flea problems all across the United States. The letters began coming a few days later. If ever anyone doubted the problems fleas cause for pets and their owners, he has only to view the stack of correspondence from persons seeking advice for eradicating, or even holding at bay, this pesky insect.

One of those who telephoned Prof. Lewis was a representative from the TONIGHT show, an NBC-TV production starring Johnny Carson. The TONIGHT show is a late-night, talk show entertainment featuring Carson and a variety of guest personalities. After a telephone interview, Prof. Lewis was invited to appear on the show. Once news of this became public, there ensued another series of interviews for local home-town newspapers, and a local TV station.

Lewis flew to Los Angeles, California, where, on June 18, he appeared with Johnny Carson on the TONIGHT show in approximately 12-15 minutes of conversation on fleas. As might be expected, emphasis was on the lighter side of the profession.

When asked about his reaction to all the publicity, Prof. Lewis admits to a certain amount of personal satisfaction from the attention but, more than that, he is pleased that through him entomology and pulicology received the national recognition they deserve - for however brief a moment. --J.H.L.

*Prof. N. Wilson, University of Northern Iowa, also was interviewed for this series on his speciality of ticks.

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Great fleas have little fleas upon their backs to bite 'em,
And little fleas have lesser fleas, and so ad infinitum.
And the great fleas themselves in turn, have greater fleas to go on;
While these again have greater still, and greater still, and so on.

Angustus de Morgan, 1872
A Budget of Paradoxes

WALL STREET JOURNAL

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THURSDAY, MAY 13, 1982

Like Fleabag Hotels, World's Flea Experts Have Image Problem

* * *

**And Pulicologist Robert Lewis
Has Bugs in a Computer;
The Rothschild Collection**

By SUE SHELLNBARGER

Staff Reporter of THE WALL STREET JOURNAL

AMES, Iowa—Not far from the bell tower on the green, rolling campus of Iowa State University is a hotbed of scientific research.

In his tiny laboratory in a university science building, Robert E. Lewis is surrounded by that scourge of dog and man—fleas. Files of 50,000 dead fleas line walls adorned by pictures of fleas. Fleas arrive in the mail. Thousands of scholarly articles devoted to fleas fill the library. In fact, Mr. Lewis personally has identified about 50 of the 2,250-odd flea species known to man, primarily by examining the genitalia of male fleas.

Those achievements make Mr. Lewis, an amiable 52-year-old professor of entomology whose graying mustache curls into neatly waxed tips, one of the world's leading experts on fleas. And his laboratory has become a clearinghouse for research by the world's flea buffs—all 150 of them. "Not very many people work on this particular insect on a world-wide basis, and no one to the extent that we do," he says.

Still, Mr. Lewis and an assistant, his wife, Joanne, keep up with research on the pesky flea and publish lists of new work in *Flea News*, a scholarly semiannual newspaper for pulicologists—flea experts. The unpretentious, stapled bulletin is sent free of charge to 150 loyal readers in 39 nations.

Fleas' Diseases

Flea News subscribers take the publication seriously, for where else can they find bibliographic listings of such scholarly articles as "Characteristics of the Ectoparasite Fauna of the Mouse-like Water Rodents of Water-oak Stands in the Central Part of the Byelorussian Polyes'ya"?

"There aren't many of us nuts working on these kinds of things, so we have to talk to one another," says Robert Traub, a microbiology professor at the University of Maryland and a leading expert on fleas and the diseases they transmit.

But Mr. Lewis, a trim man in bifocals, also exploits what he calls "the opportunities for levity" that his profession affords. A Mark Twain quote, "Fleas can be taught to do nearly everything that a Congressman can," embellishes *Flea News* 21. Sandwiched between scholarly listings in a bibliography in the same issue is the following citation: "Anonymous—Flea pulls 200,000 times its own weight. *Natl. Enquirer* 55 (14):32-33 (11 Nov.) (color picture)." The photograph cited shows the star of a European flea circus harnessed to a 1½-ounce toy locomotive and struggling across a man's fingernail.

Certainly, research shows that the flea can do wondrous things. Consider a 1973 article in *Scientific American* by noted British flea expert Miriam Rothschild and others. The oriental rat flea, best known for its central role in killing one of every three Europeans during the 14th Century by transmitting bubonic plague from rats to humans, also is a marathon hopper, the article says. If stimulated by the society of other fleas, it will jump as much as 100 times its length 10 times a minute for 72 hours without pause. Some fleas have been seen jumping about on Alpine snows. Others can survive for months frozen under the ice of Antarctica, according to the article. Although fleas spin and somersault as they leap into the air, they land on their feet 78% of the time, apparently by spreading their legs as they descend. (Pity the research scientist who had to calculate that statistic.) "If we were as strong as a flea, we could jump over the Empire State Building in one leap," observes Mr. Traub.

Protective Helmets?

Fleas' remarkable saltatorial ability may spring from body parts once devoted to flight. The bodies of fleas are so cluttered with tiny and complex structures as to force scientists intent on classifying them to write descriptions several pages long. Some fleas, for example, have helmets—probably not, as it may seem, to protect their little heads the 22% of the time they don't land on their feet, but rather to use in clinging to the hairs of their hosts. (Helmeted fleas have been found only in Australia and South America, flea experts point out, which supports the theory of continental drift—the idea that South America and Australia were once part of a much larger combined land mass.)

One noted flea scholar even suggested that fleas can sing. In an article entitled in part "The Song of the Flea," Dutch entomologist Franciscanus G.A.M. Smit wrote that the musical ability of fleas with certain types of back legs should be explored. The

insects may produce sound by rubbing leg parts against body parts. But it would seem not to matter to the fleas. "Fleas don't have ears," Mr. Lewis explains. They still may perceive sound; there's a debate about that.

Mr. Lewis's studies have taken a more practical turn. After earning his doctorate from the University of Illinois with a dissertation on the thoracic muscles of the flea, he and Mrs. Lewis—better known as Mike—set out to classify the fleas of the Middle East and to identify their hosts. During a stint with American University at Beirut, Lebanon, and in trips financed by the National Institutes of Health, the Office of Naval Research, and an oil company, Mr. Lewis has trotted the globe, from the ancient capital of Balkh in Afghanistan to the Syrian city of Palmyra, once an outpost of the Roman Empire.

"I'm not much for ruins, but they're good places to catch rats" and their fleas, Mr. Lewis says.

In the field, the host animal is trapped or shot and popped into a bag. Then the flea populace is quickly chloroformed and preserved in alcohol or slapped between laboratory slides. Such expeditions have led to the classification of more than 2,000 new flea species in this century. To keep them all straight, Mr. Lewis has arranged to program the university computer to answer flea questions with data from his library.

For example, a student might ask how many different types of fleas inhabit the mountain beaver, a small rodent in the Western U.S. The answer: four unusually large ones. The system still is being programmed, Mr. Lewis says. "We're trying to work the bugs out. And I don't mean the fleas."

Mr. Lewis plans to apply to a granting agency to finance the project. Alas, for him, classifying and documenting fleas increasingly is taking a back seat in financing and prestige to curing cancer and splicing genes.

In fact, "ribbing from your friends" is an occupational hazard of pulicology, Mr. Lewis says. Flea experts, like flea markets and fleabag hotels, aren't among the elite of their realm. There is also the actual risk of getting flea-bitten. Mr. Lewis has escaped the bites of his prey, despite his luxuriant gray beard, but Mrs. Lewis has been inhabited by unwanted guests in a Turkish hotel and in a Syrian market. The couple's dogs came down with a bad case of cat fleas last summer. "It was quite an embarrassment," Mr. Lewis says, neglecting to mention whether he means for him or the dogs.

Indeed, dog and cat fleas are the ones most familiar to most people. The cat flea is the more adaptable and widespread of the two, and it is the type found most often, even on dogs. The pests are most populous in warm climates because they can breed

year-round, indoors and out. In cold regions, the larval stages die in winter unless the fleas can breed indoors. Thus, Southern California, Florida and the South are most affected.

Sometimes, fleas do infest people. "If there's not a pet for them to jump on, when they're hungry they'll bite human beings," Mr. Lewis says. In California, many people have taken to spraying their lawns with insecticides to rid them of fleas. The insects can transmit tapeworm to human beings, but that doesn't occur often. One must eat the flea for that to happen.

Serious flea research has come a long way in the past century. Naturalist Nathaniel Charles Rothschild of the British banking family, was the first to describe the type of rat flea that was the major transmitter of plague to humans. When he died in 1923, he left the world's largest collection of fleas to the British Museum. His daughter, Miriam, carried on the family interest in fleas.

In 1914, scientists demonstrated how fleas transmit plague to human beings, and the disease, thanks to antibiotics, is now largely under control. It afflicts roughly 2,000 people in the world each year. Plague remains a sensitive political topic, however. (Egypt, mistaking a city for a nation, quarantined "the country of Denver" for a day in 1968 after a case of plague was reported in Colorado.) Public-health officials insist that research on fleas and the variety of afflictions they transmit, including tularemia and tapeworm, is still important.

Nevertheless, pulicologists' numbers are dwindling. "People are getting involved in more spectacular stuff like molecular biology," says Maryland professor Traub. "I could fly to the moon taxonomically and no one would care." Flea experts who retire are seldom replaced.

Still, new flea frontiers remain to be explored. Mr. Lewis is itching to document the fleas of South America, but "those guys down there are too wrapped up in politics" to support the proper flea research, he laments. "What we really need is a good outbreak of bubonic plague," he jokes, "and everybody will start paying attention to us again."

William L. Jellison, a retired Hamilton, Mont., scientist who did flea research for the government, couldn't agree more. He has amassed a great collection of fleas from animals ranging from dead grizzlies and mountain lions to skunks and squirrels. But after 20 years of trying, Mr. Jellison, a Flea News subscriber, hasn't been able to unload his collection to museums or universities. "I still have 2,000 fleas here," he says. "It's a challenge to get rid of this stuff."

A PREFATORY NOTE TO THE SPECIES LIST - In keeping with Mr. Smit's practice of numbering the taxa chronologically, the following list includes numbers for those species described in 1980 only. Readers are urged to call any omissions in these lists to our attention. Numbers will be assigned to the 1981 taxa later in 1983.

- 98001 paoshanensis Li & Yan 1980 Macrostylophora
- 98002 nanshanensis Tsai, Pan & Liu 1980 Ceratophyllus
- 98003 yunnanus Li, Xie & Liao 1980 Cratynius (Angustus)
- 98004 hebes clara Smit 1980 Mesopsylla
- 98005 rotundisinuata Li & Hsieh 1980 Hystrichopsylla
- 98006 pentagonia Liu, Liu & Zhang 1980 Megathoracipsylla
- 98007 kazbegiensis Goncharov 1980 Callopsylla
- 98008 angustidigita Wu, Wu & Tsai 1980 Genoneopsylla
- 98009 claviprocera Hsieh & Wu 1980 Genoneopsylla
- 98010 bisinuata Liu, Chang & Liu 1980 Genoneopsylla
- 98011 qilianensis Wu, Tsai & Liu 1980 Vermipsylla
- 98012 aprominus Liu, Cai & Wu 1980 Geusibia
- 98013 fimbriata Li & Hsieh 1980 Neopsylla
- 98014 elesina Li 1980 Neopsylla
- 98015 megaloba Li 1980 Neopsylla
- 98016 schismatosa Li 1980 Neopsylla
- 98017 eothenomus Li & Huang 1980 Ctenophthalmus
- 98018 breviprojiciens Li & Huang 1980 Ctenophthalmus
- 98019 cavaticus Li & Huang 1980 Typhlomyopsyllus
- 98020 spirataenius diqingensis Li, Xie & Yang 1980 Amphalius
- 98021 spirataenius qinghaiensis Li, Xie & Yang 1980 Amphalius
- 98022 luchunensis Huang 1980 Macrostylophora
- 98023 laxisinus Xie, He & Li 1980 Paraceras
- 98024 menetus Xie, Chen & Li 1980 Paraceras

* * *

PROVISIONAL LISTS OF TAXA DESCRIBED IN 1981 AND 1982--

1981

- Nearctopsylla pfitzeri Benton
- Amphipsylla kulkarnii Lewis
- Vermipsylla ibexa Zhang & Yu
- Araeopsylla faini Beaucournu
- Ctenophthalmus (Ctenophthalmus) bisoctodentatus suciuae Beaucournu & Grulich
- Macrostylophora abazhouensis Liu, Liu & Zhai
- Amphalius spirataenius badongensis Ji, Chen & Wang
- Amphalius clarus kunlunensis Yu & Wang
- Ischnopsyllus (Hexactenopsylla) shansiensis Liu, Hsing & Chen
- Frontopsylla (Frontopsylla) tuoliensis Yu & Zhang
- Nosopsyllus (Nosopsyllus) chayuenensis Wang & Liu
- Geusibia hemisphaera Liu, Chen & Liu
- Ctenophthalmus (Idiostenophthalmus) hopkinsi rahmi Beaucournu
- Ctenophthalmus (Idiostenophthalmus) elblae Beaucournu
- Lentistivalius occidentayunnanensis Li, Xie & Gong
- Stivalius laxilobulus Li, Xie & Gong
- Aviostenivalius hylomysus Li, Xie & Gong
- Ctenophthalmus (Euctenophthalmus) acutilobatus Wei, Chen & Liu
- Doratopsylla coreana sichuanensis Wei, Chen & Liu
- Callopsylla (Callopsylla) forfica Wu, Chen & Liu

(1973-1978)

1982

Ceratophyllus (Ceratophyllus) chasteli Beaucournu, Monnat & Launay
 Amalaraeus dissimilis daghestanicus Labunets
 Neopsylla stevensi sichuanyunnana Wu & Wang
 Ctenocephalides chabaudi Beaucournu & Bain
 Plocopsylla angusticeps Mahnert
 Hectopsylla gracilis Mahnert
 Nycteridopsylla intermedia Lewis & Wilson

* * *

An asterisk (*) preceding a citation indicates we have seen neither the original publication nor verified it in another source. Anything appearing in brackets ([]) after a citation indicates an annotation, e.g., the name of a taxon described.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1973 (list 13)

Frazier, C. A. - Biting insects. Arch. Dermat., Chicago 107(3): 400-402.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1974 (list 14)

Keirans, J. E. & C. M. Clifford - Ixodes (Pholeonixodes) conepati Cooley & Kohls (Acarina: Ixodidae): description of the immature stages from rock squirrels in Texas. J. med. Ent. 11(3): 367-369. [Siph.: 369]

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1975 (list 15)

Srivasta, S. P. & B. L. Wattal - Studies on the distribution of ectoparasites and their vertebrate hosts in urban, semi-urban (peripheral) and sylvatic zones of Nagpur City and neighboring areas (Maharashtra-India). Indian J. Ent. 35(4): 306-324.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1976 (list 13)

Franzmann, N.-E. - An investigation of the flea fauna in the nests of some high arctic birds. Ent. Meddr. 44(1): 28-30.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1977 (list 11)

Chaika, S. Yu. - The ultrastructure of glycocalyx of microvilli in the midgut of blood-sucking arthropods. Tsitologiya 19(11): 1221-1224.

Olsuf'ev, N. G. - Results and prospects of studying natural focality of tularemia in the USSR. Med. Parazit., Moskva 46(3): 273-282. [Translation 1540 of the Medical Zoology Dept., NAMRU 3, Cairo.]

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1978 (list 12)

*Eshuis, J. & P. Manschot - Communicable diseases. A manual for rural health workers. Afr. med. Res. Foundation Rural Hlth Ser. 7. Nairobi, Kenya. xiv+349 pp., figs.

Mehl, R. - Enquiries about pests and ectoparasites in 1972-76. Fauna, Oslo 31(4): 250-267, 5 figs. [Siph.: 254, 255]

(1978-1981)

Reitblat, A. G., N. P. Kalmykova & P. F. Emel'yanov - Feeding of gamasid mites (Parasitiformes, Gamasoidea) occurring in the little suslik nests. Zool. Zh. 57(11): 1732-1735.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1979 (list 11)

Carpenter, J. W. & C. N. Hillman - Husbandry, reproduction and veterinary care of captive ferrets. Ann. Proc. amer. Assoc. Zoo Vet. 1978: 36-47 [Siph.: 43]

Volkov, V. I., V. I. Katsko, A. M. Dolgikh & V. N. Zarubina - *Myopus schisticolor* Lill. (Rodentia) from the Myao-Chan Ridge and its ectoparasites. Vestn. Zool. 1979(2): 63-65.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1980 (list 7)

*Butler, T. et al. - Estudos sorológicos e experimentais da peste no Nordeste do Brasil. Rev. bras. Malariol. Doen. trop., Rio de Janeiro 32: 26-35.

*Christie, A. B. - Plague and Libya. Garyounis med. J., Benghazi 3(1): 65-67.

Coutinho, E. M., D. A. Mello & J. M. Barbosa - Patologia da infecção experimental de roedores domésticos com diferentes cepas de *Yersinia pestis*. Mem. Inst. Oswaldo Cruz 75(3/4): 53-62.

Geetha Devi, S. & R. S. Prasad - Influence of haemolymph protein concentration and gut distension on blood avidity of rat fleas *Xenopsylla cheopis* and *X. astia*. Proc. Indian natl. Sci. Acad. B46(6): 843-845. [Symp. on Insect Vector Biology held at Madras, Nov. 22-24, 1979]

Grzywinski, L., M. Mazurkiewicz & T. Martynowicz - The efficacy of Nervan 7 in the control of external parasites in hens in an intensive production. Medyc. wet. 36(10): 615-618.

Handa, M. C. & B. S. Taneja - External parasites in poultry. Indian Farming 30(4): 25-26.

*Lan, M.-y. & Y.-c. Mon - A fly and a flea both from swallow nest in Zhe-jiang. Zool. Res., Kunming 1(1): 139.

Mehl, R. - Flea bites on humans. Tidsskr. Nor. Laegeforen 100(29) IX, XI: 58.

Mehl, R. - Parasites from animals, insects and mites that attack man. Tidsskr. Nor. Laegeforen 100(10) VII, IX, XI, XIII, XV: 55-56.

Messner, B. - Massenansammlung von Flöhen. Ent. Ber. 1980(2): 90.

*Ott, A. K., A. D. Charters & R. A. Bowman - Tungiasis: imported disease. Med. J. Australia 2(11): 623-624.

*Phalke, P. G. - Ectoparasites of poultry and their control. Agriculture & Agro-industries J., Bombay 13(9): 13-15.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1981 (list 4)

Anonymous - In memoriam: Alfredo Barrera Marín (1926-1980). Folia ent. Mex. 49: 5-6.

Anonymous - Publicaciones sobre temas zoológicos de Alfred Barrera. Folia ent. Mex. 49: 11-14.

*Anonymous - Status of plague in the Americas, 1970-1980. Epidem. Bull. 2(3): 5-8.

(1981)

- *Beck, J. W. & J. E. Davies - Medical parasitology. 3rd Edition. C. V. Mosby Co., St. Louis, 355 pp., 399 figs.
- Chantral, C. - Les parasites des chiens de Rabat (Maroc). Recherches personnelles par autopsies. 1978 Univ. de Toulouse, F. R. Diss. Abstr. intl. C, 42(3): 531.
- Coman, B. J., E. H. Jones & M. A. Driesen - Helminth parasites and arthropods of feral cats. Austr. vet. J. 57(7): 324-327.
- Ejezie, G. D. - The parasitic diseases of school children in Lagos State, Nigeria. Acta Trop. 38(1): 79-84.
- Fei, R.-z., J.-y. Liu, B.-j. Xu, X. Cui & B.-z. Zhang - On the spatial distribution of the fleas in burrows of *Citellus dauricus* in Inner Mongol[ia], China. Acta ent. Sinica 24(4): 397-402.
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It will be understood that - although their titles are given in English in these lists - papers in Russian or Chinese periodicals, or books published in the USSR or China, are in Russian or Chinese (but often with summaries in English).

As we go to press, we receive, with thanks, a reprint from D. K. Mardon which adds another citation to list 2 for 1982 and another new taxon for the year.

Mardon, D. K. & F. R. Allison - A new subspecies of *Bibikovana rainbowi* (Rothschild) (Siphonaptera: Pygiopsyllidae) with records of other fleas from Queensland. J. Aust. ent. Soc. 21: 183-187. [*Bibikovana rainbowi inusitata*]

In addition to numerous telephone calls and letters in response to the WALL STREET JOURNAL article, the following poem was received. It was written by William J. Brown, M.D., of Wellesley Hills, Massachusetts, and he has kindly granted permission to reprint it in FLEA NEWS.

FLEAS

To study pulicology
You first must find a willing flea,
But fleas don't willingly enlist
To aid a pulicologist.

Devising flea taxonomy
Requires attention to detail;
The crucial factor seems to be
The sex equipment of the male.

There must be dire necessity
To prompt such curious travail.
Such tactics used on you or me
Would seem a bit beyond the pale.

To undergo such scrutiny
Is really not my cup of tea.
When probings such as these prevail
I'm glad I'm me and not a flea.

The flea inhabits many lands.
He's found, in varied size and form,
From snowy peaks to desert sands,
But favors places where it's warm.

He has no wings with which to fly
And when he walks he barely creeps.
Important moves are chiefly by
Prodigious catapulting leaps.

He rolls and tumbles through the air;
He's apt to land most any place.
And where he lands he doesn't care;
He sometimes lands upon his face.

He jumps again, and after that
He jumps again with all his might
Until he finds a dog or cat
Of whom he takes a juicy bite.

The flea in shrouded mystery
Has spread the dread bubonic plague
And changed the course of history
As much as Kissinger or Haig.

They say a few of them can sing;
 But then, of course, they have no ears.
 It seems a quite preposterous thing!
 What good is music no one hears?

This singing business lacks the ring
 Of truth. To me it all appears
 Some overzealous ding-a-ling
 Has soaked up one too many beers.

If you need fleas identified
 I hope you'll use this little guide.
 If I were you, however, I'd
 Employ a strong insecticide.

---W.J.B.

* * *

NOTES

DR. VERNON TIPTON, of Brigham Young University, Provo, Utah, recently indicated that he is anticipating early retirement from his position of Professor in the Zoology Department. Dr. Tipton joined the BYU staff in 1968 upon his retirement from the Medical Service Corps of the United States Army as Lt. Colonel.

* * *

Your attention is called to publication of the most recent supplement (Suppl. 23) to the Index-Catalogue of Medical and Veterinary Zoology, Part 5, Parasite-Subject Catalogue: Parasites: Arthropoda and Miscellaneous Phyla. July 1982, 301 pp. The Index-Catalogue is prepared by the Agricultural Research Service, U.S. Department of Agriculture. [U.S. Government Printing Office: 1982 0 - 378 - 710]

* * *

With this issue go our very best wishes for the holiday season and the new year.

Season's Greetings

Felices Fiestas

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Поздравлю

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Frohes Festtage

Meilleurs Voeux